MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 8532W 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Triode, miniature, grounded grid, RF amplifier

Outline --- 5-2 (EIA)

Base --- E7-1

Envelope --- T5-1/2

Cathode --- Coated unipotential

Base connections:

Pin No. ---	1	2	3	4	5	6	7
Element ---	g	k	h	h	g	g	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rk	Rg	Ik	Pp	TE	Alt
Unit:	V	V dc	V dc	v	Ohms	Meg	mA dc	W	°C	ft
Maximum:	6.6	150	0,-55	100	---	0.25	20	2.5	120	See note 1
Minimum:	6.0	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	150	---	---	100	---	---	---	---	---

GENERAL:

Qualification - Required

1/ Formerly type 8532

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Insp level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 1								
1301	Heater current			0.4	II	If	375	425	mA
1336	Heater-cathode leakage			0.4	II	Ihk	---	10	µA dc
1266	Total grid current	2	Rg = 0.25 Meg; Ebb = 175 V dc; Rk = 150 ohms	0.4	II	Ic	0	-0.5	µA dc
1256	Electrode current (1) (anode)			0.4	II	Ib	9.0	18	mA dc
1256	Electrode current (2) (anode)		Ec = -15 V dc	0.4	II	Ib	---	60	µA dc
1306	Transconductance (1)			0.4	II	Sm	8,800	13,200	µmhos
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes			2.5	I	R	500	---	Meg
1306	Transconductance (2)		Ef = 5.7 V	2.5	I	ΔSm Ef	---	15	%
1316	Amplification factor			6.5	S3	Mu	40	65	---
---	Noise figure	3		4.0	See note 4	NF	---	6.3	dB
1031	High-frequency vibration		Rp = 2,000 ohms	6.5	Code H	Ep	---	150	mV ac
1031	Sweep-frequency vibration		Rp = 2,000 ohms; F = 50 to 500 Hz	6.5	See note 4	Ep	---	100	mV ac
1331	Direct-interelectrode capacitance		Shield No. 316 tied to ground	6.5	Code E	Cpk	---	0.20	pF
			Shield No. 316 tied to grid			Ck-g+h	5.5	9.5	pF
			Shield No. 316 tied to grid			Cp-g+h	---	5.0	pF
			Shield No. 316 tied to ground			Chk	1.0	4.5	pF
			Shield No. 316 tied to ground			Cgp	2.3	3.3	pF
1041	Shock	5	450 G; Ehk = +100 V dc; Ec = -1.5 V dc; Rk = 0	6.5	See note 4	---	---	---	---
1031	Vibration-fatigue		5 G	6.5	See note 4	---	---	---	---

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Insp level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 2 - Continued								
---	Shock and vibration-fatigue-test end points								
1031	High-frequency vibration		R_p = 2,000 ohms	---	---	Ep	---	300	mV ac
1336	Heater-cathode leakage			---	---	Ihk	---	20	µA dc
1306	Change in transconductance (1) of individual tubes			---	---	ΔSmt	---	20	%
1266	Total grid current		Rg = 0.25 Meg; Ebb = 175 V dc; Rk = 150 ohms	---	---	Ic	0	-1.0	µA dc
1121	Base strain	6		---	---	---	---	---	---
2126	Envelope strain			2.5	I	---	---	---	---
1105	Permanence of marking			---	---	---	---	---	---

Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.0 V; Ehk = +100 V dc; Eb = Ec = 0; 1 minute "on", 4 minutes "off"	---	---	---	---
---	Heater-cycling life-test end point						
1336	Heater-cathode leakage			Ihk	---	20	µA dc
1516	Stability life		Ehk = +100 V dc; Rg = 0.25 Meg; TA = room	---	---	---	---
---	Stability life-test end point (2 and 20 hours)						
1306	Change in transconductance (1) of individual tubes			ΔSmt	---	10	%
1501	Intermittent life	7,8	Ehk = +100 V dc; Rg = 0.25 Meg; TE = 120°C (min)	---	---	---	---

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Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 3 - Continued						
---	Intermittent life-test end points (1,000 hours)						
1266	Total grid current		Rg = 0.25 Meg; Ebb = 175 V dc; Rk = 150 ohms	Ic	0	-2.0	µA dc
1301	Heater current			If	375	455	mA dc
1306	Change in transconductance (1) of individual tubes			ΔSm t	---	25	%
1336	Heater-cathode leakage			Ihk	---	20	µA dc
1211	Insulation of electrodes			R	200	---	Meg
1306	Transconductance (1), average change			Avg ΔSm t	---	15	%

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. This test shall be performed at the conclusion of the holding period.
3. Test as shown in figures 1 and 2. The 200-MHz amplifier shall have the following characteristics: (1) an overall gain sufficient to properly drive the noise figure meter (-60 to -10 dBm) is necessary for the Hewlett-Packard Model 342A, or equivalent; (2) an overall noise figure of 5.5 to 6.0 dB; (3) a minimum bandwidth of 1 MHz. The gain of the test amplifier is approximately 15 dB and no correction is made for noise contributed by the 200-MHz amplifier. Noise figure is read directly from the noise figure meter in dB.
4. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code E, shall apply.
5. A grid resistor of 0.1 megohm shall be added, except when a thyratron-type short indicator is used.
6. Acceptance sampling procedure shall be in accordance with method 1121, except that acceptance and rejection criteria shall be modified as follows:
   a. Accepted if not more than one defective for class A, B, or C defects, respectively, or if not more than a total of two defectives are found in the sample.
   b. Rejected if two or more defectives for class A, B, or C defects, respectively, or if a total of three or more defectives are found in the sample.
7. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples and shall not exceed a total of 3 tube failures.

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NOTES - Continued

8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

9. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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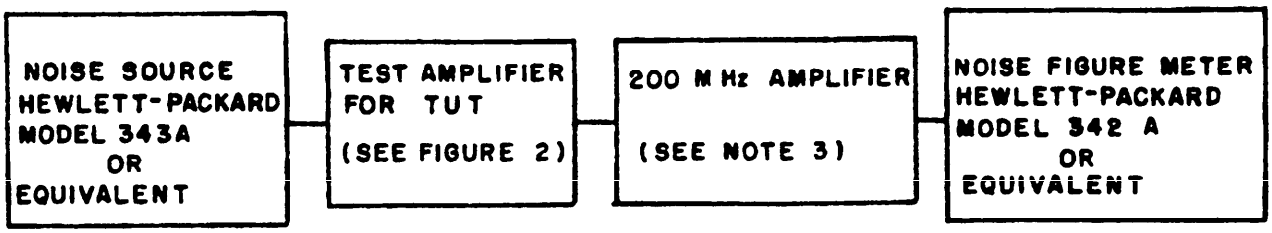
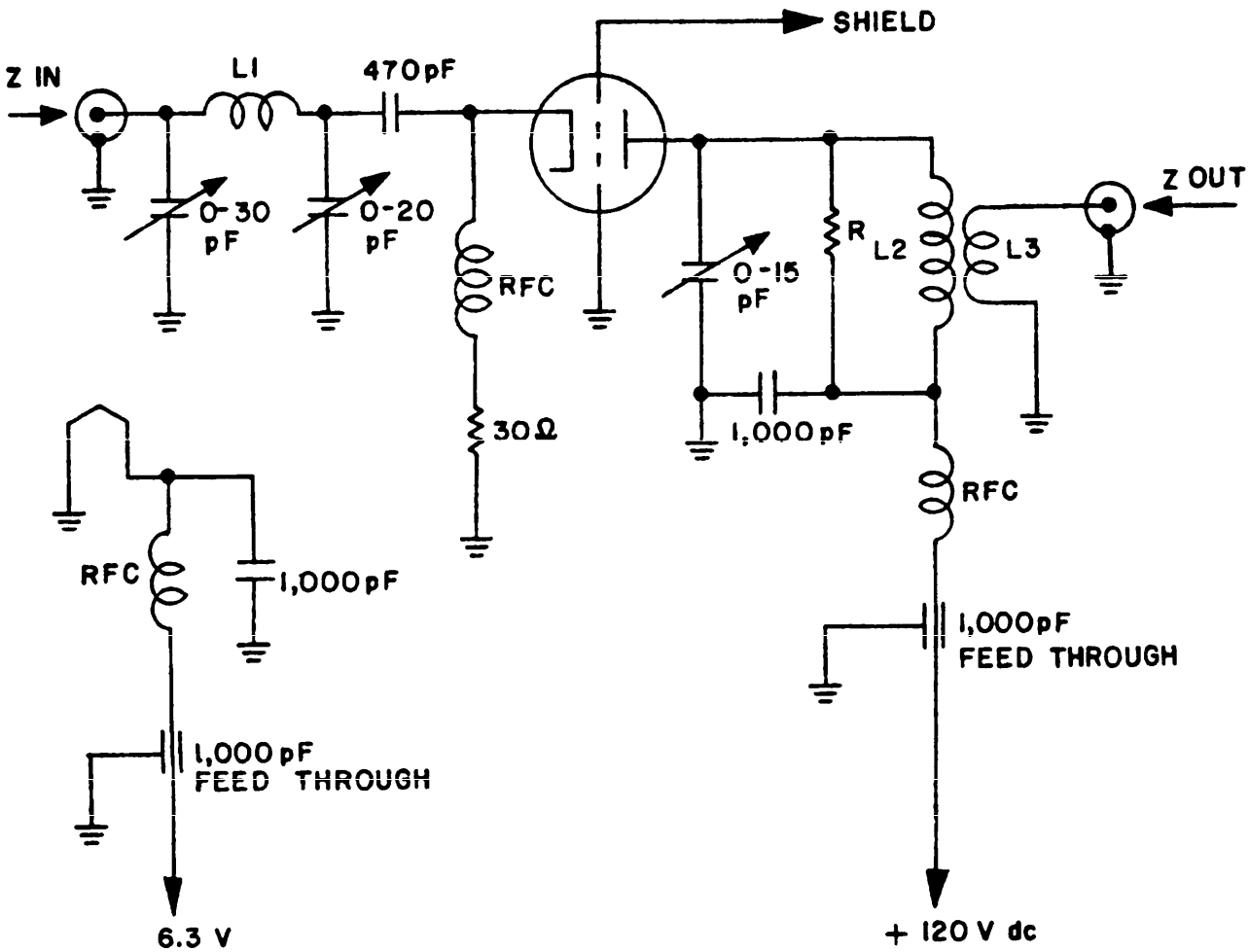


FIGURE 1. Block diagram for noise figure test.



L1 = 1 semielliptical loop No. 18 wire approximately 2 inches long.
L2 = 3 turns NO. 26 wire, 1/4 inch O.D.
L3 = 2 turns No. 26 wire, 1/4 inch O.D.
R = 5 K to 10 K chosen for optimum bandwidth (approximately 5 MHz).
RFC = .84 to 1.2 uH (Ohmite Z235 or equivalent).
Zin = 50 Ohms or matched to output of noise source.
Zout = 50 Ohms or matched to input of 200 MHz amplifier.

FIGURE 2. Test amplifier circuit for noise figure test.

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MIL-E-1/1527C Electron Tube, Receiving Type 8532W 1/

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